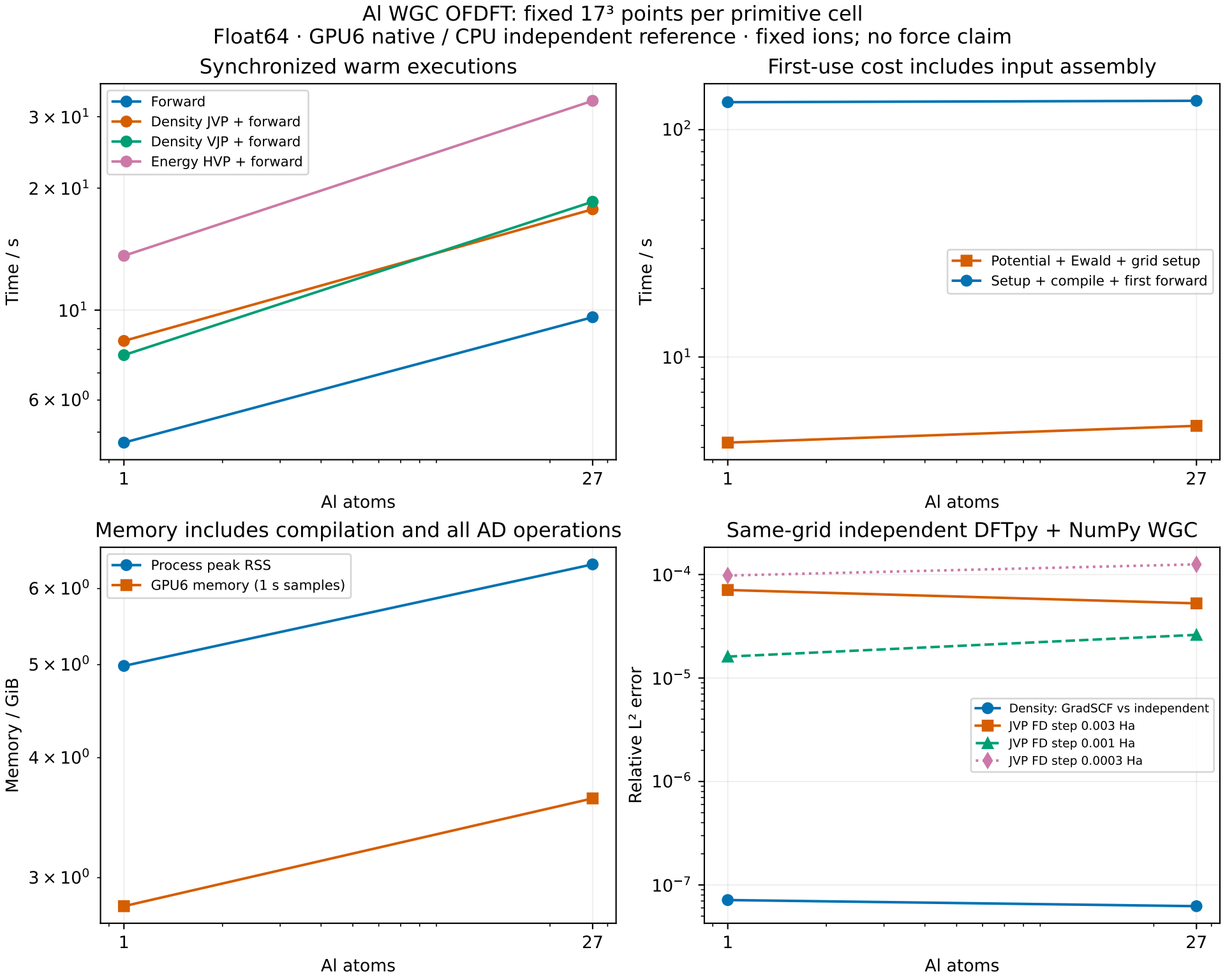
Al WGC OFDFT: fixed 17³ points per primitive cell
Float64 · GPU6 native / CPU independent reference · fixed ions; no force claim
Synchronized warm executions
Forward
Density JVP + forward
Density VJP + forward
Energy HVP + forward
Time / s
Al atoms
First-use cost includes input assembly
Potential + Ewald + grid setup
Setup + compile + first forward
Time / s
Al atoms
Memory includes compilation and all AD operations
Process peak RSS
GPU6 memory (1 s samples)
Memory / GiB
Al atoms
Same-grid independent DFTpy + NumPy WGC
Density: GradSCF vs independent
JVP FD step 0.003 Ha
JVP FD step 0.001 Ha
JVP FD step 0.0003 Ha
Relative L² error
Al atoms